Date: August 23, 2026

Prof. Kenneth M. Merz, Jr.
Editor-in-Chief
Journal of Chemical Information and Modeling
American Chemical Society

Subject: Submission of Original Research Article to *Journal of Chemical Information and Modeling*

Dear Prof. Merz and Editorial Board Members,

We are pleased to submit our original research manuscript entitled **"Physics-Gated Interpretable Machine Learning and Symbolic Regression for Double Perovskites: Overcoming Theoretical Limits Beyond 0D Compositional Models"** for consideration as a full Research Article in the *Journal of Chemical Information and Modeling* (JCIM).

Significance and Core Advances of the Work: Predicting the ground-state properties of complex multinary materials like double perovskites ($A_2BB'O_6$) using purely compositional (0D) descriptors is a major challenge in chemical informatics. While 3D graph neural networks (CGNNs) offer high fidelity, they require computationally expensive, pre-relaxed DFT coordinates and function as uninterpretable black-boxes. Conversely, foundational 0D symbolic models face theoretical accuracy ceilings ($R^2_{\text{limit}} \approx 65\%$ for formation energy ΔE_f, 60% for magnetization M, 50% for band gap E_g, and 25% for energy above hull E_{hull}) and collapse on zero-inflation point-mass densities.

In this study, we introduce a novel **Physics-Gated Interpretable Machine Learning Architecture** that breaks these theoretical literature ceilings using 100% 0D descriptors. Our key contributions include:

1. **Physics-Gated Domain Routers**: Resolving zero-inflation and derivative discontinuities at metallic/insulating and magnetic phase boundaries via high-margin hurdle models ($C = 200.0$) and soft-sigmoidal gating functions.
2. **Quantum & Thermodynamic Engines**: Deriving closed-form Harrison tight-binding interatomic matrix elements and single-perovskite tie-line proxies ($D_{\text{hull\_proxy}}$) to boost 0D accuracy without leaking 3D coordinate data.
3. **Surpassing SOTA Literature Limits**: Achieving **109.62%** of the theoretical limit for formation energy (ΔE_f $R^2 = 71.26\%$), **103.72%** for total magnetization (M $R^2 = 62.23\%$), and **101.42%** for band gap (E_g $R^2 = 50.71\%$).
4. **Rigorous Multi-Seed Validation**: Validated across 25 independent random seeds and out-of-distribution splits on 2,000 and 5,000 double perovskite compositions.
5. **Closed-Form Analytical Equations**: Discovering human-interpretable symbolic equations directly applicable for prospective materials screening.

Fit for the *Journal of Chemical Information and Modeling*: This work directly aligns with JCIM's core scope at the intersection of chemical informatics, data-driven materials modeling, and interpretable physical machine learning.

Declarations:

- This manuscript is original, has not been published previously, and is not under consideration elsewhere.
- All authors have approved the manuscript and agree to its submission to JCIM.
- All datasets and Python source code are openly available on GitHub at https://github.com/nihal-gazi/double_perovskite_compositional_interpretable.
- The authors declare no competing financial interests.

Suggested Independent Reviewers:

1. **Prof. Runhai Ouyang** (Shanghai University / FHI Berlin), *Email:* rouyang@shu.edu.cn – Expert in symbolic regression and SISSO materials informatics.
2. **Prof. Shyue Ping Ong** (UC San Diego), *Email:* ongsp@ucsd.edu – Expert in materials machine learning and perovskite stability.
3. **Prof. Chris Wolverton** (Northwestern University), *Email:* c-wolverton@northwestern.edu – Expert in DFT databases and perovskite thermodynamics.
4. **Prof. Aron Walsh** (Imperial College London), *Email:* a.walsh@imperial.ac.uk – Expert in perovskite electronic structure and tight-binding theory.

Thank you very much for your time and consideration.

Sincerely yours,

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